

Health Data Analysis – Theory Answers

1. **Null Hypothesis (H_0) and Alternative Hypothesis (H_1)**

Null hypothesis means there is no relationship or no effect between variables. It assumes everything is normal.

Alternative hypothesis means there is a relationship or effect between variables. It is opposite of null hypothesis. These hypotheses are used in statistical testing.

2. **Confidence Interval**

Confidence interval is a range of values that likely contains the true value of a population. It gives an idea about the accuracy of data.

For example, 95% confidence means we are 95% sure about the result.

3. **T-Test**

T-test is a statistical method used to compare the means of two groups. It helps to check whether the difference between groups is significant or not.

In this dataset, it is used to compare smokers and non-smokers.

4. **Chi-Square Test**

Chi-square test is used to check the relationship between two categorical variables. It shows whether variables are dependent or independent.

In this case, it is used to analyze smoking and diabetes.

5. **ANOVA Test**

ANOVA is used to compare the means of more than two groups. It helps to find if at least one group is different from others.

Here, it is used to compare age groups with glucose levels.

6. **Correlation**

Correlation shows the relationship between two variables. Its value lies between -1 and +1.

It helps to understand whether variables increase or decrease together.

7. **Data Visualization**

Data visualization means representing data using charts and graphs. It makes data easy to understand and analyze.

Common examples are bar charts, histograms, and line graphs.

8. **Mean and Standard Deviation**

Mean is the average value of the dataset. Standard deviation shows how much the data values are spread.

It helps to understand the variation in data.